Lecture 9. Classification II

Thresholds, confusion matrix, precision and recall, ROC and AUC, separability and regularized logistic regression

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9.1 Logistic regression as a classifier

Motivation. Thresholds turn probabilities into decisions; judging them needs more than accuracy, especially for rare classes.

Logistic regression predicts $P(y = 1 \mid x) = \operatorname{expit}(x^\mathsf{T}\beta)$, is fitted with cross-entropy loss, and classifies $y = 1$ when $\hat{p} \geq 0.5$.

A coin example. Tosses $0, 0, 0, 1, 1, 0, 0, 0, 1, 1$: $4 \log \theta + 6 \log(1-\theta)$ has derivative $4/\theta - 6/(1-\theta) = 0$ at $\theta = 0.4$, the sample mean (minimum cross-entropy).

The null model. (Slides: census, 24% high earners.) `ISLR2::Default`, split 80/20 as in Lecture 8. $P(y = 1) = \operatorname{expit}(\beta_0)$ gives everyone $\bar{\pi} < 0.5$, so all are classified $y = 0$.

```
set.seed(364)
idx <- sample(nrow(Default), 0.8 * nrow(Default))
train <- Default[idx, ]; test <- Default[-idx, ]
train %>% count(default) %>% mutate(pct = n / sum(n))
```

```
##   default    n     pct
## 1      No 7750 0.96875
## 2     Yes  250 0.03125
```

9.1.1 Thresholding

The decision rule need not use 0.5:

$$\text{classify}(x) = \begin{cases} 1, & \hat{P}(y = 1 \mid x) \geq t, \\ 0, & \text{otherwise.} \end{cases}$$

Raising t: fewer false positives, more false negatives.

9.1.2 Accuracy

Accuracy: proportion classified correctly. Null, `balance` only, all predictors:

```
fit_null <- glm(default ~ 1, family = binomial, data = train)
fit_1 <- glm(default ~ balance, family = binomial, data = train)
fit_all <- glm(default ~ student + balance + income, family = binomial, data = train)
classify <- function(fit, data, t = 0.5)
  factor(ifelse(predict(fit, data, type = "response") >= t, "Yes", "No"), levels = c("No
    ", "Yes"))
map_dbl(list(null = fit_null, bal = fit_1, all = fit_all),
        ~ mean(classify(.x, train) == train$default))
```

```
##     null      bal      all
## 0.968750 0.974125 0.974875
```

Small gains: the null model is already 97% accurate.

9.2 Evaluating classifiers

9.2.1 Drawbacks of accuracy

Labelling 100 e-mails (2 spam) all as ham is 98% accurate and useless.

9.2.2 The confusion matrix

Call class 1 positive and class 0 negative.

	predicted 1	predicted 0
actual 1	TP (true positive)	FN (false negative: missed)
actual 0	FP (false positive: false alarm)	TN (true negative)

$$\text{accuracy} = \frac{TP + TN}{n}, \qquad \text{precision} = \frac{TP}{TP + FP}, \qquad \text{recall} = \frac{TP}{TP + FN}.$$

Precision: share of predicted 1s that are 1; **recall**: share of actual 1s predicted 1. All-ham filter: precision $0/0$, recall $0/2 = 0$.

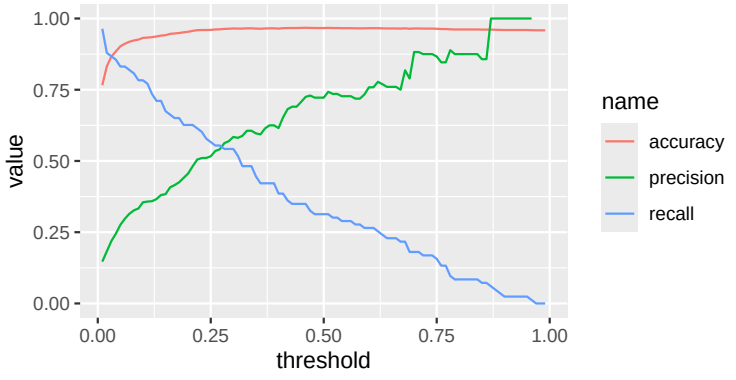
```
table(predicted = classify(fit_1, test), actual = test$default)
```

```
##          actual
## predicted   No  Yes
##       No  1907   58
##       Yes   10   25
```

The trade-off. Higher thresholds raise precision (fewer FP), lower ones recall (fewer FN; $t = 0$ gives recall 1). Accuracy peaks in between. Disease screening favours recall; a life sentence favours precision.

Other names: **sensitivity** = recall = true positive rate (TPR); **specificity** = true negative rate $= TN/(TN + FP)$; 1−specificity = false positive rate (FPR); precision = positive predictive value (PPV).

```
p_test <- predict(fit_all, test, type = "response")
y_test <- test$default == "Yes"
tibble(t = seq(0.01, 0.99, by = 0.01)) %>%
  mutate(accuracy = map_dbl(t, ~ mean((p_test >= .x) == y_test)),
         precision = map_dbl(t, ~ sum(p_test >= .x & y_test) / sum(p_test >= .x)),
         recall = map_dbl(t, ~ sum(p_test >= .x & y_test) / sum(y_test))) %>%
  pivot_longer(-t) %>%
  ggplot(aes(t, value, colour = name)) + geom_line(na.rm = TRUE) + labs(x = "threshold")
```



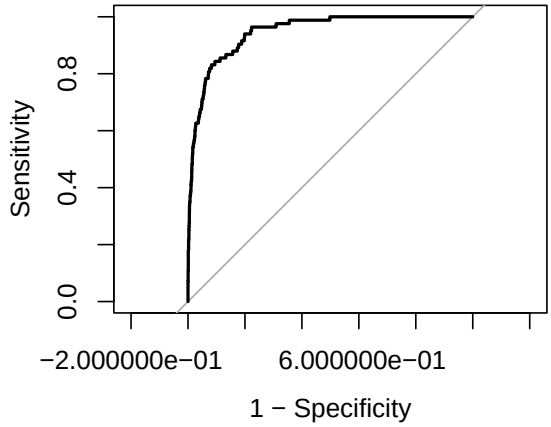
9.2.3 ROC and precision–recall curves

Over all thresholds: precision against recall is the **PR curve**; TPR against FPR is the **ROC curve**, best near the top-left. **AUC**: 1 perfect, 0.5 random. PR is more informative for unbalanced classes.

```
roc_1 <- roc(test$default, predict(fit_1, test, type = "response"), quiet = TRUE)
roc_all <- roc(test$default, p_test, quiet = TRUE)
c(auc_balance = auc(roc_1), auc_all = auc(roc_all))
```

```
## auc_balance     auc_all
##   0.9468359   0.9467919
```

```
plot(roc_all, legacy.axes = TRUE)
```



What AUC measures [Supplement]

AUC equals the probability that a randomly chosen positive case receives a higher predicted probability than a randomly chosen negative case (Mann–Whitney; Exercise 9.1): ranking quality, free of threshold and class proportions.

9.2.4 Training versus testing

Slides, census data:

model	accuracy (train)	accuracy (test)	sensitivity (test)	specificity (test)
all predictors	0.852	0.849	0.598	0.928
capital gain only	0.800	0.803	0.204	0.991
null	0.759	0.761	0.000	1.000

Test accuracy is slightly lower; the full model trades a little specificity for much sensitivity. For `Default`:

```
eval_model <- function(fit) {
  pr <- classify(fit, test)
  c(acc_train = mean(classify(fit, train) == train$default),
    acc_test = mean(pr == test$default),
    sens = mean(pr[test$default == "Yes"] == "Yes"),
    spec = mean(pr[test$default == "No"] == "No"))
}
round(rbind(all = eval_model(fit_all), balance = eval_model(fit_1), null = eval_model(
    fit_null)), 3)
```

```
##         acc_train acc_test  sens  spec
## all         0.975    0.967 0.313 0.995
## balance     0.974    0.966 0.301 0.995
## null        0.969    0.959 0.000 1.000
```

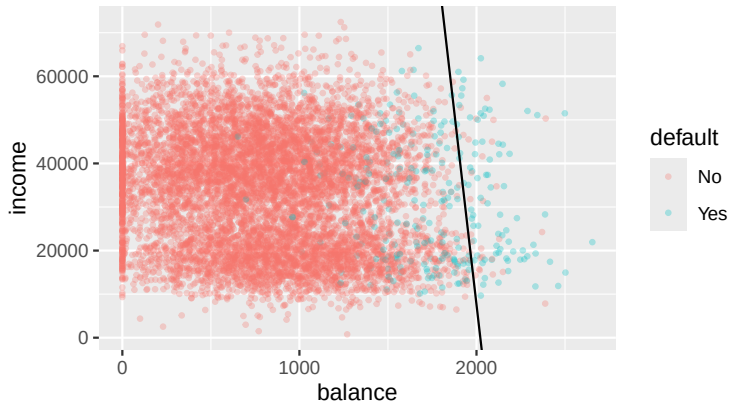
At $t = 0.5$ most defaulters are missed; lower t if misses are costly.

9.3 Linear separability and regularization

Motivation. Under perfect separation the ML coefficients diverge; regularization fixes this.

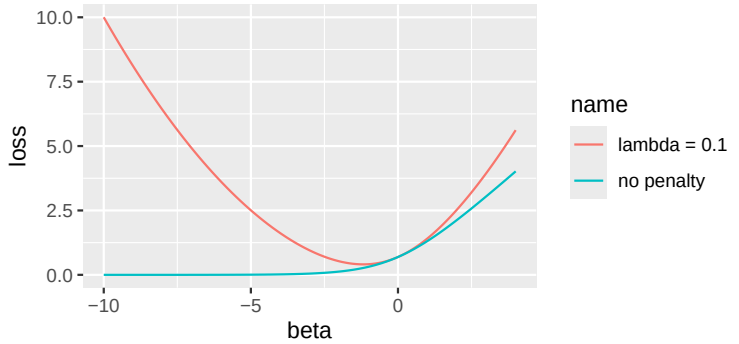
Decision boundaries. Slides: $\operatorname{logit}\hat{\pi} = -3.085 + 0.046 \cdot \text{hours}$, boundary about 67 hours; with $\hat{\pi} = \operatorname{expit}(-4.85 + 0.047 \cdot \text{hours} + 0.042 \cdot \text{age})$ it is a line in the (hours, age) plane.

```
fit_2 <- glm(default ~ balance + income, family = binomial, data = train)
b <- coef(fit_2)
ggplot(train, aes(balance, income, colour = default)) +
  geom_point(alpha = 0.3, size = 0.8) +
  geom_abline(intercept = -b[1] / b[3], slope = -b[2] / b[3])
```



Perfect separation. Points $(-1, 1)$, $(1, 0)$, model $\operatorname{expit}(\beta_1 x)$: the loss decreases to 0 as $\beta_1 \to -\infty$, predicting exactly 0 or 1; a new $(0.5, 1)$ would have *infinite* loss.

```
x <- c(-1, 1); y <- c(1, 0); expit <- function(z) 1 / (1 + exp(-z))
ce <- function(b, lambda = 0)
  -mean(y * log(expit(b * x)) + (1 - y) * log(1 - expit(b * x))) + lambda * b^2
tibble(beta = seq(-10, 4, by = 0.05)) %>%
  mutate(`no penalty` = map_dbl(beta, ce), `lambda = 0.1` = map_dbl(beta, ce, lambda =
    0.1)) %>%
  pivot_longer(-beta) %>%
  ggplot(aes(beta, value, colour = name)) + geom_line() + labs(y = "loss")
```



Linearly separable: a hyperplane separates the classes; coefficients diverge (`glm()` warns of fitted probabilities 0 or 1).

Regularized logistic regression adds a penalty, e.g. L_2:

$$-\frac{1}{n}\sum_{i=1}^{n}\left\{y_i \log \operatorname{expit}(x_i^\top \beta) + (1-y_i)\log\left(1-\operatorname{expit}(x_i^\top \beta)\right)\right\} + \lambda \sum_{j=1}^{p} \beta_j^2.$$

It has a finite minimum even under separation (Exercise 9.3). Standardise, then `glmnet(..., family = "binomial")`:

```
X <- scale(model.matrix(default ~ student + balance + income, data = train)[, -1])
cv_fit <- cv.glmnet(X, train$default, family = "binomial", alpha = 0)
coef(cv_fit, s = "lambda.min")
```

```
## 4 x 1 sparse Matrix of class "dgCMatrix"
##                       s1
## (Intercept) -4.83965221
## studentYes  -0.11306543
## balance      1.81973290
## income       0.01566196
```

9.4 Selected exercises

Motivation. Proofs of three claims of the lecture.

Exercise 9.1 (AUC as a ranking probability)

A classifier gives scores $s_1, \dots, s_m$ to the positive cases and $t_1, \dots, t_k$ to the negative cases, all $m + k$ scores distinct. Show that the area under the ROC curve equals

$$\mathrm{AUC} = \frac{1}{mk} \sum_{i=1}^{m} \sum_{j=1}^{k} \mathbb{1}(s_i > t_j),$$

the probability that a randomly chosen positive case scores higher than a randomly chosen negative case.

Solution [Supplement].

Step 1. At threshold c the classifier has $\mathrm{TPR}(c) = \#\{i : s_i \geq c\}/m$ and $\mathrm{FPR}(c) = \#\{j : t_j \geq c\}/k$. Lowering c moves $(\mathrm{FPR}, \mathrm{TPR})$ from $(0, 0)$ to $(1, 1)$, up $1/m$ at each positive score and right $1/k$ at each negative one, one at a time (distinct scores): a staircase.

Step 2. The area is the sum over horizontal steps of width times height: the step at t_j has width $1/k$ and height $\#\{i : s_i > t_j\}/m$.

Step 3. Summing over j gives the formula: the share of the mk (positive, negative) pairs ordered correctly. (Ties give diagonal segments, counting $\frac{1}{2}$.) ■

Exercise 9.2 (precision depends on prevalence)

A test has sensitivity se and specificity sp, and the positive class has prevalence $\pi = P(y = 1)$. Show that

$$\text{precision} = P(y = 1 \mid \hat{y} = 1) = \frac{\mathrm{se}\,\pi}{\mathrm{se}\,\pi + (1 - \mathrm{sp})(1 - \pi)}.$$

Evaluate it for $\mathrm{se} = \mathrm{sp} = 0.99$ with $\pi = 0.5$ and with $\pi = 0.001$, and explain why the ROC curve but not the precision–recall curve is unaffected by the class proportions.

Solution [Supplement].

Step 1. By definition $\mathrm{se} = P(\hat{y} = 1 \mid y = 1)$ and $1 - \mathrm{sp} = P(\hat{y} = 1 \mid y = 0)$. By the law of total probability $P(\hat{y} = 1) = \mathrm{se}\,\pi + (1 - \mathrm{sp})(1 - \pi)$, and Bayes' theorem gives the formula.

Step 2. $\pi = 0.5$: 0.99. $\pi = 0.001$: $0.00099/(0.00099 + 0.00999) = 0.090$, nine in ten positives are false alarms.

Step 3. Sensitivity and specificity condition on the true class, so the ROC curve does not depend on π; precision mixes the classes through π, so the PR curve shifts with prevalence. ■

Exercise 9.3 (separation and the ridge penalty)

Let $L(\beta)$ be the mean cross-entropy loss of logistic regression for data (x_i, y_i), where x_i includes the constant 1 and all coefficients are penalised.
(a) Suppose some b satisfies $b^{\mathsf{T}} x_i > 0$ whenever $y_i = 1$ and $b^{\mathsf{T}} x_i < 0$ whenever $y_i = 0$. Show that L has no minimiser.
(b) Show that for every $\lambda > 0$ the penalised loss $L_\lambda(\beta) = L(\beta) + \lambda\|\beta\|^2$ has exactly one minimiser, whether or not the data are separable.

Solution [Supplement].

(a) Each term of L is $\log(1+e^{-\eta_i})$ if $y_i = 1$ and $\log(1+e^{\eta_i})$ if $y_i = 0$, with $\eta_i = \beta^{\mathsf{T}} x_i$; both are strictly positive, so $L(\beta) > 0$ for every finite β. Along $\beta = tb$ every $y_i = 1$ term is $\log(1 + e^{-tb^{\mathsf{T}} x_i}) \to 0$ and every $y_i = 0$ term is $\log(1 + e^{tb^{\mathsf{T}} x_i}) \to 0$ as $t \to \infty$. Hence $\inf L = 0$, and the infimum is not attained.

(b) *Existence.* L_λ is continuous and $L_\lambda(\beta) \geq \lambda\|\beta\|^2$ because $L \geq 0$. $K = \{\beta : L_\lambda(\beta) \leq L_\lambda(0)\}$ is non-empty, closed and inside $\|\beta\|^2 \leq L_\lambda(0)/\lambda$, hence compact; the minimum on K is global since $L_\lambda > L_\lambda(0)$ outside K.

Uniqueness. L is convex (its Hessian $\frac{1}{n}\sum_i q_i(1-q_i)x_i x_i^{\mathsf{T}}$ is positive semidefinite, Lecture 8) and $\lambda\|\beta\|^2$ is strictly convex, so L_λ is strictly convex. If $\beta \neq \beta'$ were both minimisers with value v, then $L_\lambda(\frac{\beta+\beta'}{2}) < \frac{1}{2}L_\lambda(\beta) + \frac{1}{2}L_\lambda(\beta') = v$, a contradiction. ■